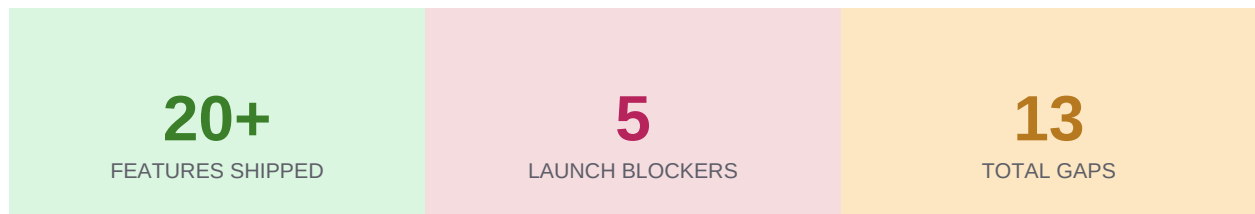


PRODUCTION READINESS AUDIT

Phase 1 launch readiness review.

Code-level audit of the NammaSign Flutter app and admin web portal — every feature except the payment flow (deferred by design).



Prepared for JERBO DAVID BLESSON

Audit scope excludes payment flow (Razorpay) by user direction.

Code base: /Users/jerbo/Desktop/nammasign_phase1

Executive summary

NammaSign has shipped 200+ tracked engineering items and the core marketplace flow is complete end-to-end. The product is functional and the code is well-organised. Three categories block a public launch: a hardcoded admin credential in the deployed web bundle, missing Apple App Store privacy strings, and zero crash / analytics instrumentation. Each is fixable in under a day.

Below is the full picture across nine readiness dimensions.

Dimension	Status	Notes
Core feature scope	READY	20+ features shipped & wired to live Firestore
Security – Firestore rules	READY	Role gates, scoped writes, server-only mutations
Security – Storage rules	READY	Per-uid scoping, 5/10/50 MB caps aligned with client
Security – PII	READY	Aadhaar last-4 only; admin doc-based gating
Security – admin auth	BLOCKER	Dev creds hardcoded in client (admin_login_screen.dart:21)
Error handling	READY	Firebase init guard, humanised auth errors
Observability	BLOCKER	No Crashlytics, Analytics, or Performance Monitoring
Testing	GAP	Effectively zero automated tests
iOS App Store readiness	BLOCKER	Missing NSCameraUsage / NSPhotoLibraryUsage strings
Android Play Store ready	GAP	R8 minify off; namespace still com.example.*
Compliance (DPDPA / GDPR)	BLOCKER	No Privacy Policy or Terms links anywhere
UX – stub tiles	GAP	5 'Phase 1b' snack stubs still visible to users
Performance	GAP	No image caching; no pagination on history/listings
CI / CD	GAP	Manual builds; no GitHub Actions / Codemagic
Payments	DEFERRED	Out of scope by explicit user choice

Launch blockers

FIX FIRST

These five items must be resolved before the first public release. Total estimated effort: 1–2 dev-days plus legal copy review.

1

Hardcoded admin credentials in client bundle

The admin login screen ships with the email and password literally embedded as Dart constants. After flutter build web, both values land in the public JavaScript bundle on <https://namma-empire.web.app>. Anyone with the URL can view-source and sign in.

Fix: Delete the dev defaults; require the admin to type credentials. Optionally pull from build-time env or a vault.

```
lib/admin/features/auth/screens/admin_login_screen.dart:21-26
```

2

Missing iOS privacy strings

Info.plist has no NSCameraUsageDescription, NSPhotoLibraryUsageDescription, or NSPhotoLibraryAddUsageDescription. The app uses file_selector + video_player + KYC uploads from the camera roll. Apple will reject submission.

Fix: Add ~3 short usage strings (1–2 lines each) to Info.plist. If you ever add analytics SDKs, also wire App Tracking Transparency.

```
ios/Runner/Info.plist (needs NSCameraUsage... keys)
```

3

Zero crash / analytics instrumentation

No firebase_crashlytics, firebase_analytics, or firebase_performance in pubspec.yaml. FlutterError.onError only writes to the console; in production every crash is invisible and you can never know how customers actually use the app.

Fix: Add firebase_crashlytics + firebase_analytics; route FlutterError.onError and PlatformDispatcher.instance.onError to Crashlytics; log key user events.

4

No Privacy Policy or Terms of Service anywhere

grep -rin 'privacy|terms' lib/ returns nothing. Login, signup, and profile have no links. India's DPDPA 2023 requires explicit consent, a Privacy Notice, and grievance officer contact info. Google Play and Apple App Store also require both.

Fix: Host two URLs (Privacy Policy + Terms); add links on login/signup screens; add a consent checkbox or 'By continuing you agree...' affordance.

5**User-visible 'Phase 1b' stubs still wired**

Five tiles still show snack messages like 'Apple sign-in coming in Phase 1b'. Users will tap and feel cheated.

Affected:

- login_screen.dart:194 — Apple sign-in button
- payment_failure_screen.dart:288 — Support tile
- profile_screen.dart:117 — KYC documents tile
- campaign_status_screen.dart:122 — Campaign menu
- premium_tab.dart — Premium tab teaser

Fix: Hide each entry-point if the destination isn't built, or finish the destination.

What's already production-grade

READY

Twenty-plus shipped features run against real Firebase. Each entry below has been audited at code level — security rules, error paths, and edge cases are in good shape.

Marketplace flow

- ✓ Phone OTP + Google sign-in with humanised error messages
- ✓ Account-type wizard (Corporate / Individual) gated by router
- ✓ Corporate + Individual signup with KYC file upload to Storage
- ✓ Live area listings with search; 3-step booking wizard
- ✓ Booking persistence with serverTimestamp; status-driven history view
- ✓ Campaign status screen (Under Review / Live / Not Approved variants)
- ✓ Account deletion (Cloud Function + dialog + cascade to Storage)

Admin web portal

- ✓ Auth gate via admins/{uid} doc existence
- ✓ Bookings queue with stale-alert (>1 hour pending) blink + tile
- ✓ Booking detail with Mark paid / Approve / Reject actions
- ✓ Users list + detail with KYC verify/reject + Danger Zone delete
- ✓ Areas list + Add area dialog (Firestore writes)
- ✓ Finance: GST/tax dashboard, 12-month breakdown, CSV export
- ✓ Help & FAQs CMS with publish/draft toggle

Cross-cutting

- ✓ FCM push (Android + iOS) with token rotation + dead-token pruning
- ✓ In-app notifications bell with live unread badge
- ✓ Light / Dark / System theme toggle, persisted via SharedPreferences
- ✓ Tax invoice PDF generator + native share sheet
- ✓ Cloud Functions: status-change push, hourly campaign sweep, account deletion
- ✓ Firestore + Storage rules with proper list/get split, size caps aligned to client

Other gaps (post-blocker)

FIX SOON

These are not launch-blockers but should land in the first one or two post-launch sprints.

6

Testing — effectively zero

test/widget_test.dart is a placeholder. No unit tests for repositories, validators, or notifiers. No integration_test/ directory.

Suggested first cuts: golden tests for BookingCard and OrderSummaryCard, unit tests for FinanceStats.from(), BookingTotals.compute(), PickedFile.validate().

7

Android — R8 / ProGuard disabled

isMinifyEnabled = false. Larger APK, no obfuscation, slower startup. Acknowledged in code comments but should be flipped before Play Store.

[android/app/build.gradle.kts:64-66](#)

8

Performance — no image caching

Bare Image.network used for creatives and KYC previews. Every list scroll re-downloads the same images. Add cached_network_image with shimmer placeholders.

[admin/features/bookings/screens/booking_detail_screen.dart:189](#)

9

Performance — no Firestore pagination

userBookingsStreamProvider streams ALL of a user's bookings; same for listings. For a power user this scales linearly. Add .limit(50) + load-more.

[features/history/data/repositories/bookings_repository.dart:54](#)

10

Wrong CFBundleName + namespace

iOS CFBundleName is still 'nammasign_phase1'; Android namespace still 'com.example.nammasign_phase1'. applicationId is correct. Cosmetic but worth fixing before App Store screenshots.

[ios/Runner/Info.plist:18](#) • [android/app/build.gradle.kts:23](#)

11

Documentation drift

README still says 'Phase 1a — UI only. Auth runs against an in-memory fake repository'. Entire app shipped with real Firebase and admin portal since then.

[README.md](#)

12

No CI / CD

Builds are manual. Adds risk and slows release cadence. Add GitHub Actions or Codemagic with: dart analyze, build APK on tag, Firebase Hosting deploy on main.

13

Single environment

All builds point to the production namma-empire Firebase project. No staging. Add a second Firebase project + flutterfire configure --project staging.

Nice-to-have (defer)

OPTIONAL

- Branded native splash screen via flutter_native_splash
- Per-event FCM notification channels for fine-grained user control
- Convert remaining ListView calls to ListView.builder
- Enable uploadDebugSymbols in firebase.json once Crashlytics ships
- Document a release-notes structure in /docs
- Reverse-DNS namespace cleanup (com.example.* → com.nammaempire.*)
- Notifications screen pagination at 60 rows

Deferred — Payment flow

BY DESIGN

Razorpay payment integration was explicitly removed by your direction. Bookings land as **pending_review** and the admin team contacts customers offline to collect payment. The Cloud Functions, Flutter wiring, and Razorpay package are all gone. The only residue:

- A '256-bit encryption · Powered by Razorpay' label on the review screen (payment_review screen) that should be removed for honesty.
- The 'pending_payment' status in BookingStatus enum is unused today but cheap to keep.

Recommended 5-day plan to ship

ACTION

Day	Focus
Day 1	Remove hardcoded admin creds; add iOS privacy strings; flip R8; fix CFBundleName + namespace; rewrite README.m
Day 2	Install Crashlytics + Analytics; wire FlutterError.onError + PlatformDispatcher.onError to record; instrument 8–10 key eve
Day 3	Draft & host Privacy Policy + Terms (legal copy review); add in-app links + consent affordance; add data-deletion web U
Day 4	Hide or finish the 5 'Phase 1b' stub tiles; add cached_network_image; add .limit() + load-more on history + listings.
Day 5	Set up GitHub Actions (lint + APK build on tag + Hosting deploy on main); add first 10 unit/golden tests; submit App Stor